Can recently logged sites support the recolonization and accessible harvest of hardy native and culturally important species? A proposal for the study of *common yarrow* (*Achillea Millefolium* L.) in the Okanagan region of BC.

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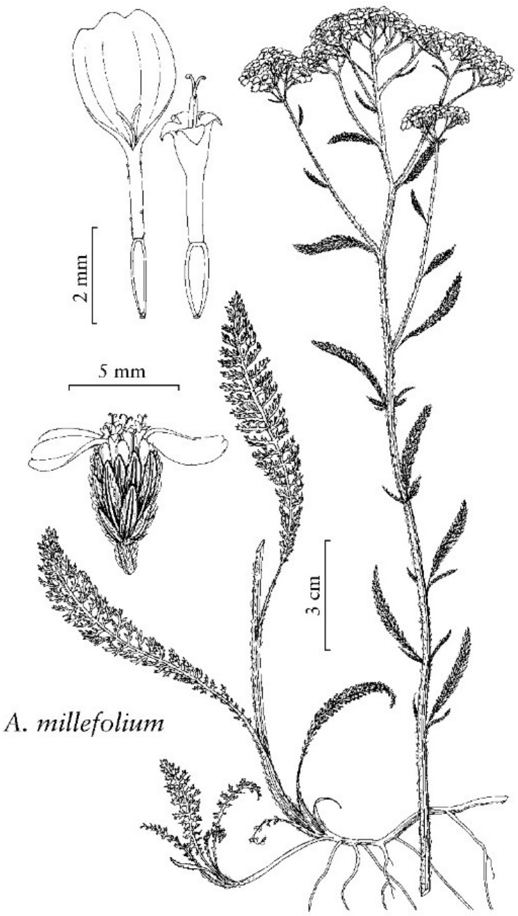
Introduction

Common yarrow (*Achillea Millefolium* L) is an herbaceous perennial found across the temperate regions of North America, Europe, and Asia. Several ecogeographic variants (vars. *alpicola lanulosa, borealis,* and *pacifica*) are found across British Columbia (BC)'s coastal meadows, lower alpine, and interior arid steppe (E-Flora BC, 2020; Hebda, 2017). Yarrow is frequently found along roadsides, where it benefits from full sun, tolerates poor soils, does not require watering, withstands mowing, and contributes to erosion control (Burton & Burton, 2003, p. 108-110). As a medicinal plant used by BC's First Nations for aches and pains as well as commercially in pharmaceutical products (BC Invasives, 2023; Burton & Burton, 2003, p. 110), recolonization of recently disturbed areas by yarrow presents an opportunity to provide competition against invasive plants (eg. Scotch broom, *Cytisus scoparius*) and provide accessible harvesting sites for users of yarrow.



Illustrator: Linny Heagy, in *FNA Volume 19.*

Background

Yarrow has been widely studied via surveying and field plots, where its species-specific niche has been identified in terms of elevation, slope and aspect, and nutrient regimes (Klinkenberg, 2013). Since yarrow is often found along roadsides, sites of recent and semi-frequent ongoing disturbance from the clearing of trees and understory, it can be hypothesized that the felling of the tree and shrub layers creates a meadow patch which allows yarrow and similar hardy native herbs to thrive. This study, therefore, will assess the distance between yarrow patches and disturbance, as well as the time from disturbance event to determine if yarrow preferentially establishes spatially and/or temporally nearby disturbed areas. Malcolm Knapp Research Forest (MKRF) is an ideal site as it both meets the climatic conditions for yarrow potential (per Klinkenberg, 2013) and has a well-documented history of disturbance locations and dates.

Methods
First GIS will be used for plot selection. From the MKRF geodatabase, looking at the land with non-forest vegetation cover type, disturbances will be classified as area which have been harvested within the past 10 years or are within 10m of roads, trails, or buildings. Because we want just the vegetated area, this will be identified with a "donut buffer" for these latter elements (ie. create a hole for the feature, leaving just the 10 meter range outside of the feature. Next a stratified sample of plots will be taken – 5 within each disturbance type (cutblock, road, trail, building). Field work will occur during flower blooming season (June-September), where the point sampled will be the centroid of a 2x2m study plot. At each plot the approximate percent cover of common yarrow within the plot will be recorded. Back with spatial data, the distance (in m) from disturbance, time since disturbance (in years), and percent cover of yarrow will be joined into one dataset. Boxplots will be generated to visualize this difference, and t-tests will be performed to determine if the results deviate from the expected distribution (ie. are skewed) in a statistically significant way.

Expected Results and Significance
A boxplot and t-test for statistical significance will be used to assess whether the sample *does* or *does not* display a trend towards occupying sites spatially or temporally near disturbances. If our results indicate a significant skew towards the disturbances (ie, a), we can confirm our hypothesis that common yarrow prefers colonizing disturbed landscapes. If the skew is indicated for a landcover class or BEC zone, we can suggest that yarrow is well-established in that region of BC and suggest its planting after human disturbances occur to facilitate access to these culturally important plants and reduce the risk of colonization by invasive and destructive species.

References

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